

Introduction to Software Engineering

Software engineering is a technique through which we can develop or create software for computer systems and any other electronic devices.

In other words, software engineering is a process in which user needs are analyzed and software is designed based on these needs.

In SE the development of software using well-defined scientific principles, methods and procedures.

Software engineers build these software and applications by using designing & programming languages.

In order to create a complex S/W we should use S/W engineering techniques as well as to reduce the complexity we should use abstraction and decomposition where, abstraction describes only the important part of the S/W and leave irrelevant things for later stages of development. So, the requirement of S/W becomes simple. And decomposition breaks down the S/W into a number of modules where each module produces a well-defined independent task.

Purpose of sw engineering

- 1} To manage large software. (Simple)
- 2} For greater scalability.
- 3} To manage the cost.
- 4} To manage the dynamic nature of the sw.
- 5} For better quality management.

large sw - required requirement
 [2000 LOC] gathered
 - Analyzed requirement
 - create or ready design
 then start next work

⇒ If client yes of project coding.

⇒ In coding we must used the abstraction & decomposition

⇒ we create different - 2 module for diff. work.

add () a+b
 sub () a-b

then client will easily understand.
 otherwise, coding or program will seems lengthy or complex.

So, we follow the principle of SE.
 As result, large software will manage easily.

Cost - Maintenance Cost

SE under project or website maintenance like error remove, updations etc. as per depend on client requirement.

Scalability - simple programming with the help of SE.
- easy to recreate the requirements.

Dynamic Nature - If we create s/w in software engineering, then the prog. nature shows dynamic. Means we can easily any thing add, remove like say maintenance in the s/w.

Better quality - we ^{need} follow the concept of software engineering.

⇒ Without SE - we can't create the s/w
⇒ If follow SE - we know that s/w cost, maintenance, coding, testing

Software Engineering

It consists of two words software & Engineering

Software - It is a collection of programs that when executed provide desired functions & performance.

Ex - MS word, MS Excel, Bank Management System, Bus etc.

⇒ Execute the desired fxn need to write a set of programs are called as s/w.

System s/w, Application s/w

Date: _____

: Engineering - The process of using knowledge & principles to design, build & analyze objects.

Defⁿ "Software engineering is the application of engineering principles to the design, development, maintenance & implementation of software."

Software product - Software when made for a specific requirement is called S/w product.

// If we develop a specific S/w for specific industry or bank etc is called a S/w product.
Like Bank S/w is a S/w product.

Software process - It is the set of activities & associated results that produce a S/w product.

// Perform those activities to produce a S/w product is called a S/w process.

OR - The process that deals with the technical & management issues of S/w development is called a S/w process.

// S/w issues resolve with the help of S/w development

Software Product Characteristics :->

- ① Maintainability
- ② Correctness
- ③ Reusability
- ④ Reliability
- ⑤ Portability
- ⑥ Efficiency

① Maintainability — It is the ease with which changes can be made to satisfy new requirements or to correct deficiencies. — problem update

— Software should be flexible enough so that future changes can be easily.

— Maintenance accounts for nearly 70% of the cost of S/w development. Therefore it is an important characteristics of S/w product.

② Correctness — It is the degree with which S/w perform its specific requirements correctly.

— At the start of S/w development, the requirements of S/w are determined & formalized in the requirements specification document.

None Create

- Various testing techniques are used for correctness of software product.

Acc. to requirement S/w satisfied or not.

③ Reusability - It is ease with which Software can be reused in developing other S/w.

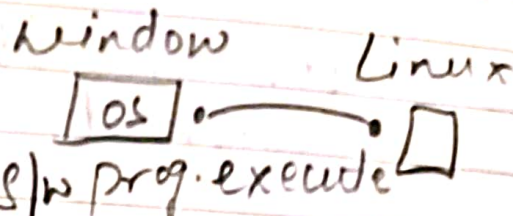
- By reusing existing S/w to create complex S/w, can reduce the development time of new S/w by developers.

① Reliability - S/w reliability is the probability of failure free S/w operation for a specified period of time in a specified environment.

- Ideally S/w engineers want their products to fail as little as possible & be as easy as possible to fix that means S/w product have high correctness & good maintainability.
- However it would be difficult to imagine a highly reliable system.

⑤ Portability - S/w portability means ability to move S/w from one m/c platform to another.

- When hardware changes, the software can also run for new S/w.



- when Computer S/w configuration changes, it is less expensive to port software on new software configuration of computer as compared to developing new S/w.

⑥ Efficiency - Efficiency refers to the ability of the S/w to use system resources in the most effective & efficient manner.

- The software should make effective use of storage space & execution time.

- But space & execution time are inversely related to each other, that is increasing the execution efficiency causes a decrease in the space efficiency.

- So S/w can not be designed with efficiency in every aspect.
- According to requirement, S/w products are designed.

S/w process ^{Activities} characteristics $\Rightarrow$

A S/w process is a set of activities & associated results whose objective is the advance development or evolution of software.

process activities

- [S/w specification
- [S/w development
- [S/w validation
- [S/w evolution

① S/w Specification - In this activity, customers & S/w engineers define the S/w to be produced & the constraints on its operations & functions. conditions

Requirement Gathered into a document

② S/w Development - Here the S/w engineer design & programmed the S/w as per customer specifications.

Coding - Designing - Implement
document form convert into Code

③ S/w validation - In this activity S/w is checked to ensure that it is what the customer requires as in the specification.

Sp. ^{Design} Dev. ^{check} Coding ^{validate} through various testing techniques.

④ S/w evolution - In this activity S/w is modified to reflect changing customer & market promote requirements.
(Including maintenance)

S/w process characteristics :-

① Understandability - To what extent is the process explicitly defined & how easy is it to understand the process definition.

// process should be easy to understand.

② Visibility - Do the process activities culminate or complete in clear results so that the progress of the process is externally visible?

// which perform the process activity of S/w & S/w progress results is visible or not.

③ Supportability — To what extent can CASE (Computer-aided S/w Engineering) tools be used to support the process activities?

④ Acceptability — Is the defined process acceptable to & usable by the engineers responsible for producing the Software product?

// S/w process activity should be easy to understand, visibility, supportability with used automatic tools & acceptable by engineers.

// Don't be complex process activities. It should be easily understood by S/w engineers.

⑤ Reliability — Is the process designed in such a way that process errors are avoided or trapped before they result in product error?

⑥ Robustness — Can the process continue in spite of unexpected problem?

⑦ Maintainability — Can the process evolve to reflect changing organizational requirements or identified process improvements?

// Can we update the requirement in the process which we apply in S/w

⑧ Rapidly - How fast can process of delivering a system from a given specification be completed?

// How fast we can complete all the process of activities of the S/W according to ^{given} specific requirements.

Imp utilize of an inappropriate S/W process may reduce the quality or worth of the S/W product &/or raise the development costs.

// S/W process must be follow all the characteristics. If process activities is an inappropriate then the results of S/W given or generate low quality results of S/W & may be raise the development or maintenance charges of S/W with the all requirements full fill.